



Recognising special number sequences

- 1 Which of these could be the first four terms in an arithmetic sequence? Tick **2** correct answers
- ☒ 11, 8, 5, 2, ...
- ☐ -7, -1, 1, 7, ...
- ☐ 27, 33, 39, 46, ...
- ☒ 118, 120, 122, 124, ...
- ☐ 5, 10, 20, 40, ...
- 2 6, 30, 150, 750, ... are the first four terms in a geometric sequence with common ratio 5. Fill in the blank
- 3 36, 49, 64, 81, ... are the first four terms in a sequence of square numbers. The next term is 100. Fill in the blank
- 4 Which of these are triangular numbers? Tick **3** correct answers
- ☒ 1
- ☒ 3
- ☒ 6
- ☐ 9
- ☐ 12
- 5 Select the expression that represents 4 more than $a + 4$. Tick **1** correct answer
- ☒ $a + 8$
- ☐ $a + 4 + a + 4$
- ☐ $4(a + 4)$
- ☐ $4a + 4$
- ☐ $4a + 8$



6 Which of these could be the first four terms in a geometric sequence? Tick **2** correct answers

☐ -1, 2, -10, 20, ...

☒ 2, -4, 8, -16, ...

☐ 3, 5, 8, 12, ...

☐ 5, 10, 15, 20, ...

☒ 10, 30, 90, 270, ...

